Dataset Description:

Link: `danielfdias98/derm-reasoning-full-reasoning` from Huggingface.

Number of rows: 5,086.

Dataset Split:

Train: 4069 (80%),

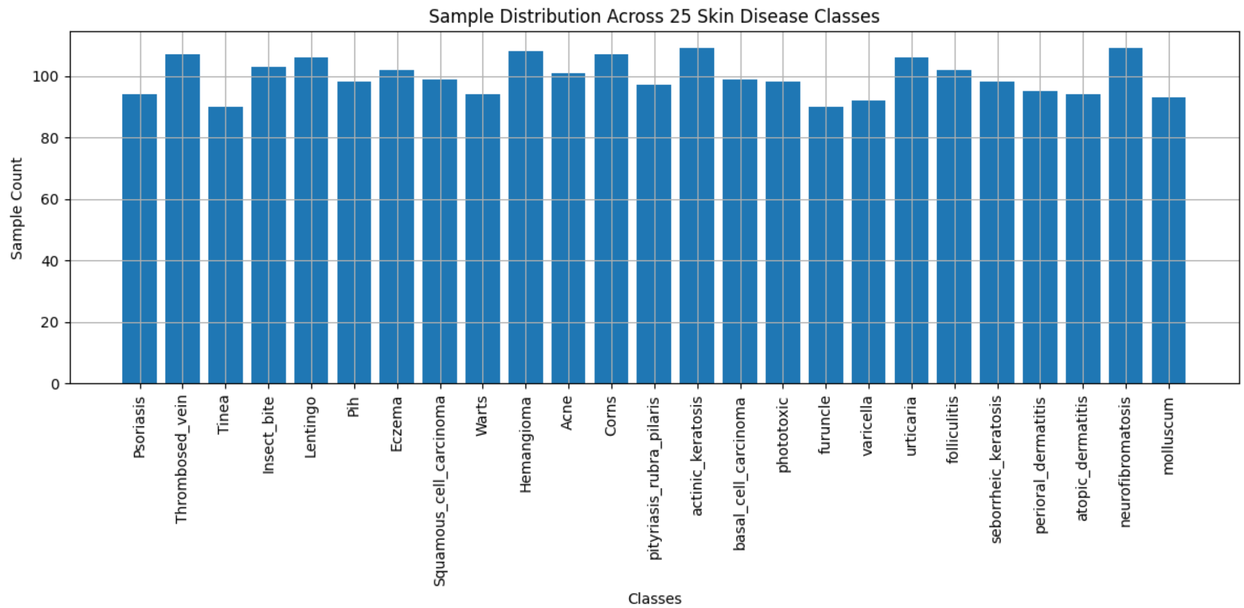
Validation: 509 (10%),

Testing: 508 (10%)

Image Dimension: 720x489

Number of Classes: There are 25 classes in this dataset.

- Psoriasis
- Thrombosed_vein
- Tinea
- Insect_bite
- Lentingo
- Pih
- Eczema
- Squamous_cell_carcinoma
- Warts
- Hemangioma
- Acne
- Corns
- pityriasis_rubra_pilaris
- actinic_keratosis
- basal_cell_carcinoma
- phototoxic
- furuncle
- varicella
- urticaria
- folliculitis
- seborrheic_keratosis
- perioral_dermatitis
- atopic_dermatitis
- neurofibromatosis
- molluscum



Model Description:

The Gemma 3N 4B (E4B) model is a parameter-efficient multimodal transformer based on the MatFormer architecture, designed for scalable deployment across edge and resource-constrained environments. It integrates a language decoder with dedicated vision (MobileNet-V5) and audio (USM-based) encoders, enabling unified processing of text, image, and audio inputs. The model employs Per-Layer Embedding (PLE) caching and selective parameter activation to reduce memory footprint while maintaining performance. Although the full model contains approximately 8 billion parameters, it operates with an effective size of 4 billion parameters, allowing efficient inference on low-resource hardware.

- **Architecture Type**: Matryoshka Transformer (MatFormer)
- **Layers**: ~35 transformer layers
- The E4B model contains smaller sub-models (E2B).
- Only a subset of parameters can be activated depending on compute budget.

Component	Details
Effective Parameters	~4B (E4B)
Actual Parameters	~8B total
With PLE	~6.8B+ active at runtime
Trainable (Our Setup)	~19M (LoRA, ~0.24%)

- Based on MobileNet-V5 backbone.
- **Output Modality**
- Text-only output.
- Inputs can be:
  - Text.
  - Image.

    - Audio.
- ~300M parameters.
- Supports resolutions:
    - 256×256
    - 512×512
    - 768×768

Model Finetuning:

The base model, Gemma 3N 4B, was fine-tuned using a parameter-efficient approach to adapt it for instruction-following tasks involving text and visual inputs. Instead of updating all model parameters, we employed Low-Rank Adaptation (LoRA) to reduce computational cost while maintaining performance. Fine-tuning was applied to both:

- Language components (text understanding and generation)
- Vision-language interaction layers (for handling image-conditioned responses)

The audio modality was not utilized, and corresponding components remained frozen during training. We used the implementation provided by Unsloth to inject LoRA adapters into the transformer architecture.

Configuration:

- Rank (r): 8
- Scaling factor (α): 8
- Dropout: 0
- Bias: None

LoRA adapters were applied to:

- Attention modules
- Feed-forward (MLP) layers
- Language backbone
- Vision-language projection layers

This resulted in training only ~19.2 million parameters, corresponding to approximately 0.24% of the total model parameters, significantly reducing memory and compute requirements.

The model was loaded in a 4-bit quantized format to enable efficient training on limited hardware.

- Quantization: 4-bit (bitsandbytes)
- Maximum sequence length: 1024
- Precision: Mixed precision (automatic dtype selection)

This configuration allowed the model to be trained on a single NVIDIA Tesla T4 GPU (≈14.7 GB VRAM), the experiments were done in Kaggle environment. Fine-tuning was performed on a subset of the FineTome-100k dataset, consisting of 3,000 samples in a conversational format.

Preprocessing steps:

- Conversion to Gemma-3 chat template format
- Standardization of conversation structure
- Removal of duplicate special tokens (e.g., <bos>)

Each training sample follows a multi-turn dialogue format, where:

- User input may include text and/or image references.
- Model output corresponds to the assistant response.

Fine-tuning was conducted using the Supervised Fine-Tuning (SFT) paradigm via Hugging Face TRL. **Training hyperparameters:**

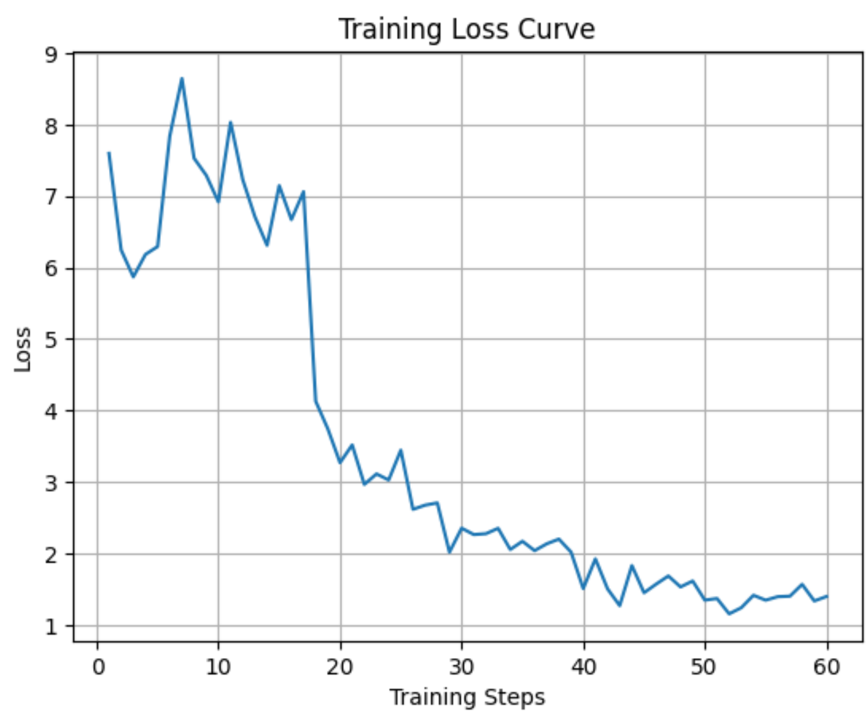
- Batch size (per device): 1
- Gradient accumulation steps: 4
- Effective batch size: 8
- Learning rate: 2×10^{-4}
- Optimizer: AdamW (8-bit paged)
- Weight decay: 0.01
- Learning rate scheduler: Linear
- Warm-up steps: 5
- Training steps: 60

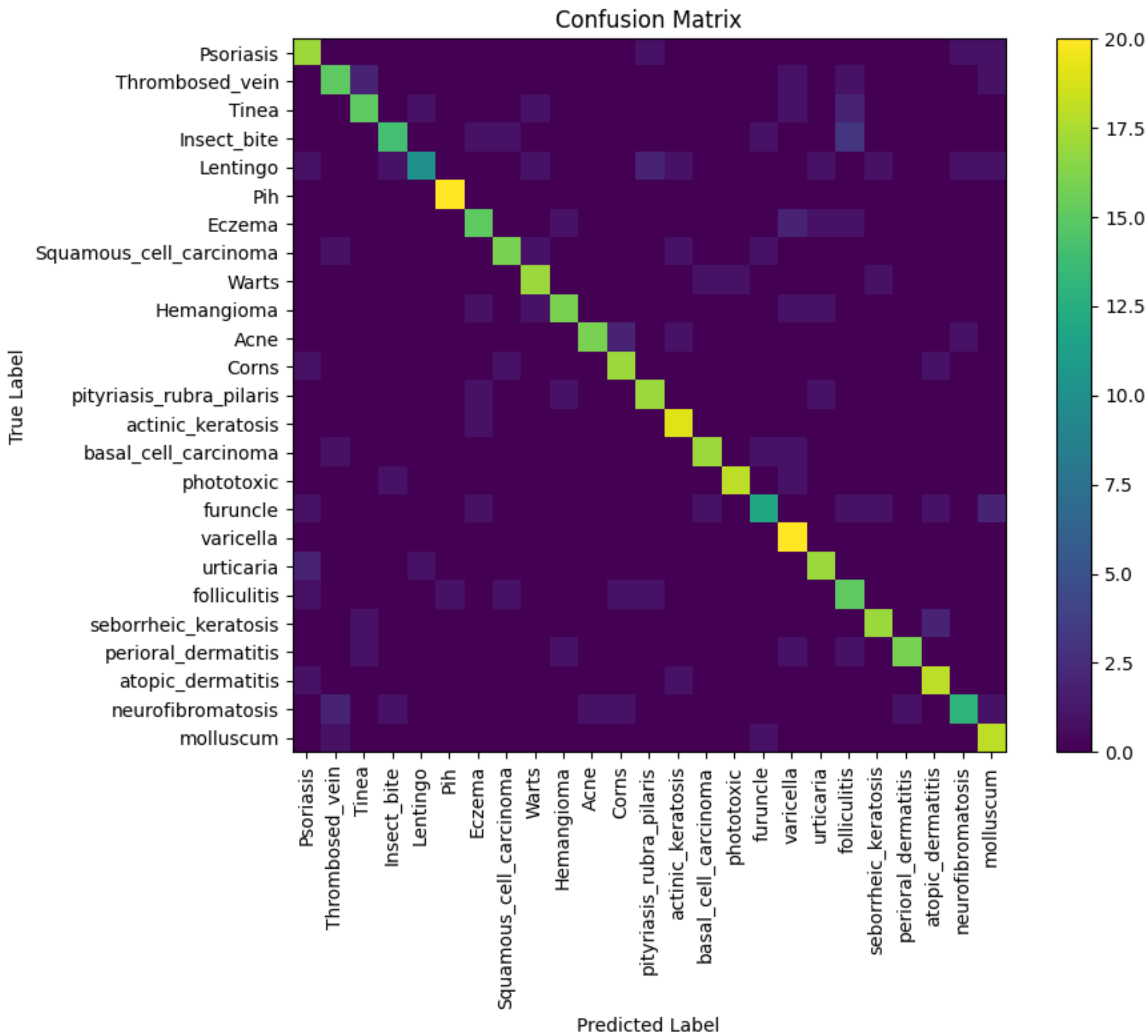
Computational Resources

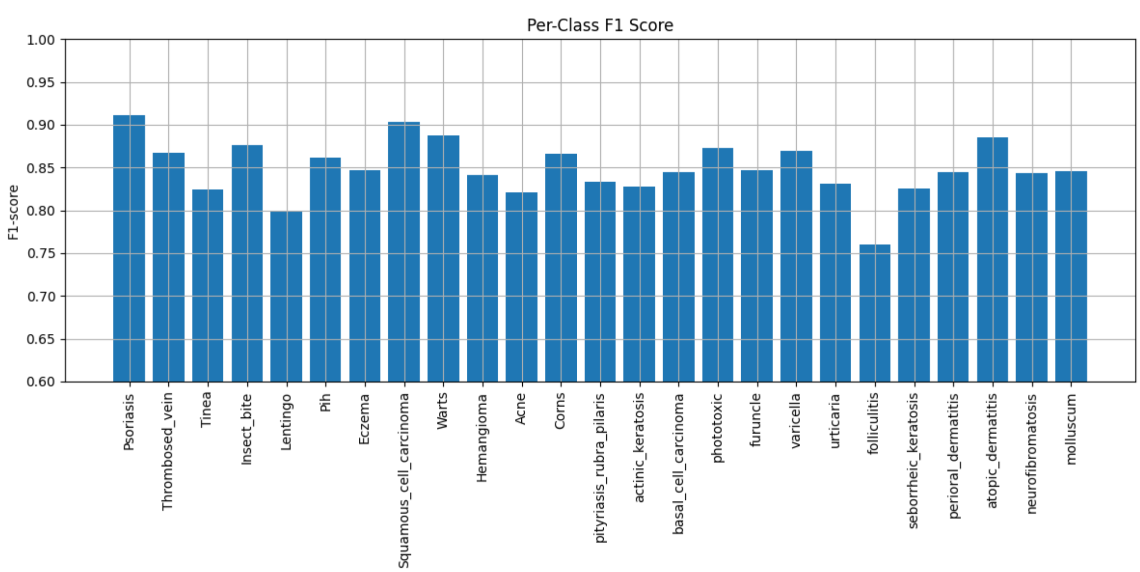
- GPU: NVIDIA Tesla T4
- Total memory: ~14.7 GB
- Peak memory usage: ~12.59 GB (~85.4%)
- Training time: ~16.3 minutes

The use of LoRA and 4-bit quantization enabled efficient training within limited computational resources.

Outputs:

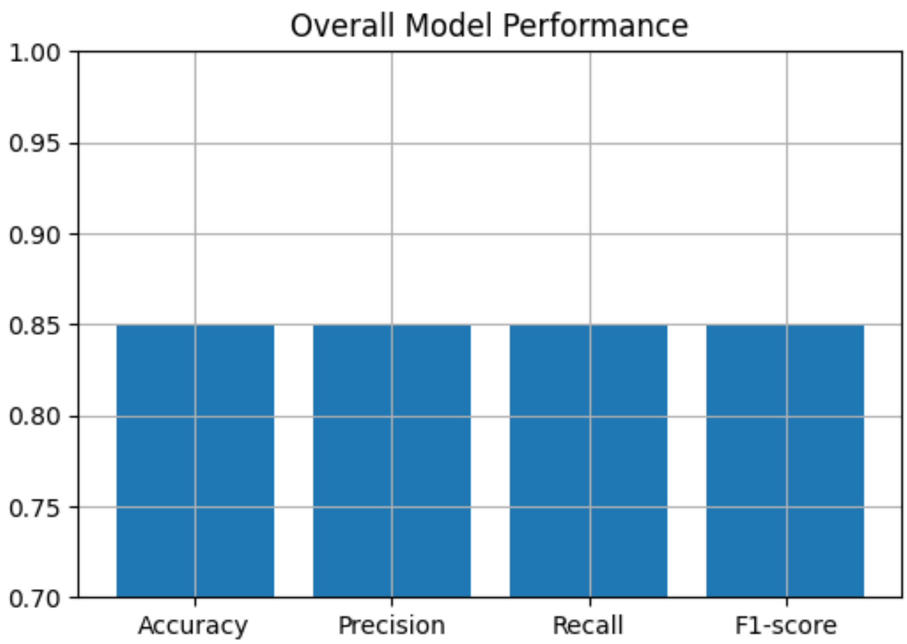



Confusion Matrix
True Label
Predicted Label
Psoriasis
Thrombosed_vein
Tinea
Insect_bite
Lentingo
Pih
Eczema
Squamous_cell_carcinoma
Warts
Hemangioma
Acne
Corns
pityriasis_rubra_pilaris
actinic_keratosis
basal_cell_carcinoma
phototoxic
furuncle
varicella
urticaria
folliculitis
seborrheic_keratosis
perioral_dermatitis
atopic_dermatitis
neurofibromatosis
molluscum
20.0
17.5
15.0
12.5
10.0
7.5
5.0
2.5
0.0


Per-Class F1 Score
F1-score
1.00
0.95
0.90
0.85
0.80
0.75
0.70
0.65
0.60
Psoriasis
Thrombosed_vein
Tinea
Insect_bite
Lentingo
Pih
Eczema
Squamous_cell_carcinoma
Warts
Hemangioma
Acne
Corns
pityriasis_rubra_pilaris
actinic_keratosis
basal_cell_carcinoma
phototoxic
furuncle
varicella
urticaria
folliculitis
seborrheic_keratosis
perioral_dermatitis
atopic_dermatitis
neurofibromatosis
molluscum

Overall Metrics

- Accuracy: 0.854
- Precision: 0.850
- Recall: 0.852
- F1-score: 0.849



References:

[1] @misc{dias2026derm-reasoning, author = {Ferreira Dias, Daniel}, title = {Dermatology Reasoning Dataset: Structured chain-of-thought annotations across five public sources}, year = {2026}, howpublished = {\url{https://huggingface.co/datasets/danielfdias98/derm-reasoning}}, }

[2] @article{gemma_3n_2025, title={Gemma 3n}, url={https://ai.google.dev/gemma/docs/gemma-3n}, publisher={Google DeepMind}, author={Gemma Team}, year={2025}}